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Fri 17 Jul 2026 16:36:23 - sadmlib_std_demo.py v03.28.00 - Library v4.69.4
Short description of program
sherlock.maison.ca - User: root - Umask: 0002 - Arch: x86_64
Redhat Linux v9.8 - Kernel 5.14.0-687.24.1.el9_8.x86_64 - SADMIN(S): /opt/sadmin
=====
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[OK] Connection to sadmin database establish.

01) SADMIN Python Library functions available to developers

[001]	sa.get_release()	SADMIN Release Number (XX.XX)	2.1.2
[002]	sa.get_ostype()	OS Type (In Uppercase,LINUX,AIX,MACOS)	LINUX
[003]	sa.get_osversion()	Return O/S Version (Ex: 7.2, 6.5)	9.8
[004]	sa.get_osmajorversion()	Return O/S Major Version (Ex 7, 6)	9
[005]	sa.get_osminorversion()	Return O/S Minor Version (Ex 2, 3)	8
[006]	sa.get_osname()	O/S Distribution (In uppercase)	REDHAT
[007]	sa.get_oscodename()	O/S Project Code Name	N/A
[008]	sa.get_username()	Current User Name	root
[009]	sa.get_kernel_version()	Return O/S Running Kernel Version	5.14.0-687.24.1.el9_8.x86_64
[010]	sa.get_kernel_bitmode()	Return O/S Kernel Bit Mode (32 or 64)	64
[011]	sa.get_hostname()	Return Host Name	sherlock
[012]	sa.get_host_ip()	Return Host Main IP Address	192.168.1.48
[013]	sa.get_domainname()	Return Host Domain Name	maison.ca
[014]	sa.get_fqdn()	Return Fully Qualified Domain Name	sherlock.maison.ca
[015]	sa.get_serial()	Return System serial Number	0
[016]	sa.get_epoch_time()	Return Current Epoch Time	1784320584
[017]	sa.epoch_to_date(1784320584)	Convert epoch time to date	2026.07.17 16:36:24
[018]	sa.date_to_epoch(wdate)	Convert Date to epoch time	1784320584
	DATE1 (End): 2026.06.30 10:00:44	DATE2 (Start): 2026.06.30 09:50:03	
[019]	sa.elapsed_time (DATE1,DATE2)	Elapse Time between two timestamps	00:10:41
[020]	sa.get_package_type()	Return package type (rpm,deb,lp, dmg)	rpm
[021]	sa.get_arch()	Get system architecture	x86_64
[022]	sa.lock_system(hostname)	Lock the specified hostname (Not FQDN)	True
[023]	sa.lock_status(hostname)	Check if host specified is lock	True
[024]	sa.unlock_system(hostname)	Unlock host specified, monitoring on	True
[025]	sa.write_log('message'[,lf=true])	Write message to Screen,Log or Both	message
[026]	sa.write_err('message'[,lf=true])	Write message to Error and Std Log	message
[027]	sa.sendmail(addr,sub,body,attach)	Send email (body=textFile) & opt.attach	0=Success 1=Error
[028]	sa.show_version(sa.ver)	Show Program & Library version & Desc.	scriptName -v
[029]	sa.sleep(60,15)	Sleep 60 sec & update every 15 sec.	60...45...30...15...0
[030]	sa.trimfile(filename,nlines=500)	Keep last 500 lines of filename.	0=Success 1=Error
[031]	sa.get_nb_cpu	Return number of CPUs	2
[032]	sa.get_nb_core_per_socket()	Return number of core per socket	2
[033]	sa.get_nb_thread_per_core()	Return number of threads per core	1
[034]	sa.get_nb_socket()	Return number of socket	1
[035]	sa.silentremove('file')	Silent file delete (no msg, no err)	0
[036]	sa.get_server_cpu_speed()	Get Maximum CPU Speed	0 MHz
[037]	sa.touch_file('filename')	Update timestamp or create file	0
[038]	sa.db_connect('sadmin')	Open connection to database	0=Success 1=Error
[039]	sa.db_close(conn,cur)	Close connection to database	0=Success 1=Error

02) Variables that influence library behavior, appears in SADMIN Code Section of your code
 We recommend, to set these variables prior to calling 'sa.sadm_start()'.

[001]	sa.ver	Program version number	03.28.00
[002]	sa.desc	Program description	Short description of program
[003]	sa.pn	Program name with extension	sadmlib_std_demo.py
[004]	sa.inst	Program name without extension	sadmlib_std_demo
[005]	sa.root_only	Program can only be run by 'root'	False
[006]	sa.server_only	Program can only run on SADMIN server	False
[007]	sa.sadmgrp_only	Run only if part of SADMIN Grp or root	False
[008]	sa.multiple_exec	Can run more than 1 copy simultaneously	False
[009]	sa.quiet	Quiet False=ErrNo+Mess True=OnlyErrNo	False
[010]	sa.log_type	Send Output to [S]creen [L]og [B]oth	B
[011]	sa.log_append	True=Append to log, False=New log	False
[012]	sa.log_header	Generate header in log	True
[013]	sa.log_footer	Generate footer in log	True
[014]	sa.use_rch	Save execution data to '.rch' file	False
[015]	debug	Debug Level 0-9 (Incr. Verbose)	0
[016]	username	Current user name	root
[017]	pid	Current Process ID	\$pid
[018]	exit_code	Program Exit Return Code	0
[019]	sa.errno	Error No. returned by function called	0
[020]	sa.errmsg	Error Mess. returned by function called	
[021]	sa.hostname	Current Host Name	sherlock
[022]	sa.sadm_pid_timeout	PID file default TimeToLive (Sec)	14400 sec
[023]	sa.sadm_lock_timeout	Maximun nb. sec. a host can be lock	7200 sec
[024]	sa.get_ostype()	OS Type (In Uppercase,LINUX,AIX,MACOS)	LINUX
[025]	sa.sadm_alert_type	0=NoMail 1=OnError 3=OnSuccess 4=All	1
[026]	sa.sadm_alert_group	Error Group (Default Group)	default
[027]	sa.sadm_warning_group	Warning Group Name	warning
[028]	sa.sadm_info_group	Info Group Name	info
[029]	sa.sadm_alert_repeat	0=AlertOnce or Sec. before alert repeat	0 sec
[030]	sa.sadm_mail_addr	SADMIN Administrator Email(s)	brucetalbot95@gmail.com
[031]	sa.sadm_max_logline	Trim log to this maximum of lines	500 lines
[032]	sa.sadm_max_rchline	Trim rch file to this max. of lines	30 lines
[033]	sa.sadm_email_startup	Send email to SysAdmin on startup	Y
[034]	sa.sadm_email_shutdown	Send email to SysAdmin on shutdown	Y
[035]	cmd_ssh_full	SSH cmd to client	/usr/bin/ssh -qnp 32
[036]	current_time	Current date & Time	2026.07.17 16:36:23
[037]	sa.db_used	Need Access to SADMIN Database	True
[038]	sa.db_name	Database Name (Defaut:'sadmin')	sadmin
[039]	sa.db_conn	Database connector	<pymysql.connections.Connection object at 0x7fe44fafb730>
[040]	sa.db_cur	Database cursor	<pymysql.cursors.DictCursor object at 0x7fe44fafb6d0>
[041]	sa.db_name	Database name	sadmin

03) Python Library - Overview of the 'start()' and 'stop()' functions.

Function 'start()'

The 'start()' function basically initialize the SADMIN environment.
When you call 'start()', it will only come back to caller, only if everything went OK,
Otherwise, it will advise the user of the error and exit(1).

- If 'db_used' is True, it will create a connection to the Database and return a connection (db_conn) and cursor (db_cur) objects.
Example: '(db_conn,db_cur) = sa.start()'
- If 'sa.db_used' is False, then it will return None for both 'sa_conn' and 'sadb_cur'.
Example: 'sa.start()'

Here a summary of the different things the 'start()' function does :

- 1) Make sure \$SADMIN directories and sub-directories exist and have proper permissions.
- 2) If 'log_append' is True, it open standard/error log in append mode, else in write mode.
- 3) If 'log_header' is True, write the log header.
- 4) If 'root_only' is True and the current user is not root, show error message and exit(1).
- 5) If 'server_only' is True and you are not on the SADMIN server, show error message and exit(1).
- 6) If 'sadmgrp_only' is 'True', then current user must be part of the SADMIN group (unless root).
- 7) If 'db_used' is True, open connection to database and return connection and cursor objects.
If not on SADMIN system, show error message and exit(1).
- 8) If lock_status() is True and running on the SADMIN server, issue message and exit(1).
If the creation of the lock was done more than 'lock_timeout' seconds, the lock file is remove.
- 9) If PID file 'pid_file' exist and 'sa.multiple_exec' is True, continue normal execution.
If PID file exist and execution time (sec) is less than the 'pid_timeout', show error mess. and exit(1).
If PID file exist and execution time (sec) exceed the 'pid_timeout' a new 'pid_file' is created and execution is resume.
- 10) If 'use_rch' is True, add a line in 'rch' file with code '2' and starting time of your program.

Function 'stop()'

This function accept one parameter, either a 0 for Success or a 1 for Error.
This should be the one of the last function called at the end of your program.

What this function does:

- 1) Calculate execution Time.
- 2) If 'use_rch = True', update the rch file (End Time & Elapse Time ...).
- 3) Validate & determine the alert group name (sadm_alert_group) use in rch file.
- 4) If 'db_used' is True and we are on the SADMIN server, close the database connection.
- 5) If the error log file (.elog) is empty, delete it.
- 6) Delete the PID file of the script (pid_file).
- 7) If 'log_footer' is True, write the log footer.
- 8) Close log and error log files.
- 9) If 'sa.max_logline' is not zero, trim the log according to user choice in 'sa.max_logline'.
- 10) If 'sa.max_rchline' is not zero, trim the 'rch' according to user choice in 'sa.max_rchline'.
- 11) Set permission and owner/group to log and rch files.
- 12) If on the SADMIN server, then rch and log are immediatly web central directory.

4) SADMIN configuration file content - List of variables available to developers.

----- General Section -----

[001]	sa.sadm_server	SADMIN server name (FQDN)	sherlock.maison.ca
[002]	sa.sadm_host_type	SADMIN [C]lient or [S]erver	S
[003]	sa.sadm_cie_name	Your Company name	The Batcave
[004]	sa.sadm_domain	Server creation default domain	maison.ca
[005]	sa.sadm_user	SADMIN User Name	sadmin
[006]	sa.sadm_group	SADMIN Group Name	sadmin
[007]	sa.sadm_pwd_random	Daily change password of '\$SADM_USER'	Y
[008]	sa.sadm_pid_timeout	PID file default TimeToLive (Sec)	14400 sec.
[009]	sa.sadm_lock_timeout	Maximun nb. sec. a host can be lock	7200 sec.

----- Database Section -----

[010]	sa.sadm_dbname	SADMIN Database Name	sadmin
[011]	sa.sadm_dbhost	SADMIN Database Host	localhost
[012]	sa.sadm_dbport	SADMIN Database Host TCP Port	3306
[013]	sa.sadm_ro_dbuser	SADMIN Database Read Only User	squery
[014]	sa.sadm_ro_dbpwd	SADMIN Database Read Only User Pwd	*Hidden*
[015]	sa.sadm_rw_dbuser	SADMIN Database Read/Write User	sadmin
[016]	sa.sadm_rw_dbpwd	SADMIN Database Read/Write User Pwd	*Hidden*

----- Web Interface Section -----

[017]	sa.sadm_www_user	User that Run the Web Server	apache
[018]	sa.sadm_www_group	Group that Run the Web Server	apache
[019]	sa.sadm_monitor_update_interval	Monitor web page refresh rate (Sec)	60 Seconds
[020]	sa.sadm_monitor_recent_count	Monitor web page nb. recent scripts	5
[021]	sa.sadm_monitor_recent_exclude	Monitor web page exclude list	sadm_nmon_watcher,sysmon_script_test

----- Email Section -----

[022]	sa.sadm_mail_addr	SADMIN SysAdmin Email(s)	brucetalbot95@gmail.com
[023]	sa.sadm_smtp_server	Your internet smtp server	smtp.gmail.com
[024]	sa.sadm_smtp_port	Your internet smtp server port	587
[025]	sa.sadm_smtp_sender	Your internet smtp email address	brucetalbot95@gmail.com
[026]	sa.sadm_gmpw	Your internet smtp email password	*Hidden*
[027]	sa.sadm_email_startup	Send email to SysAdmin on startup	Y
[028]	sa.sadm_email_shutdown	Send email to SysAdmin on shutdown	Y

----- Monitoring Section -----

[029]	sa.sadm_alert_type	0=NoAlert 1=OnError 3=OnSuccess 4=All	1
[030]	sa.sadm_alert_group	Error Group Name (Default Group)	default
[031]	sa.sadm_warning_group	Warning Group Name	warning
[032]	sa.sadm_info_group	Info Group Name	info
[033]	sa.sadm_ssh_port	Default SSH Port to talk with client	32
[034]	sa.sadm_textbelt_key	TextBelt.com API Key	*Hidden*
[035]	sa.sadm_textbelt_url	TextBelt.com API URL	https://textbelt.com/text
[036]	sa.sadm_ntfy_email	Email related to NTFY	duplessis.jacques@gmail.com
[037]	sa.sadm_ntfy_pwd	Password related to NTFY	5ne46yJFgmQS24rU8e9
[038]	sa.sadm_ntfy_token	Token related to NTFY	tk_2a511lu2o46k5hdgt64h5u55wt0
[039]	sa.sadm_ntfy_topic	Topic use with NTFY	sadmdev
[040]	sa.sadm_ntfy_url	URL to send notification to NTFY	https://ntfy.sh

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[041] sa.sadm_ntfy_user          NTFY user name          duplessis.jacques@gmail.com

----- Files and Logs pruning Section -----
[042] sa.sadm_nmon_keepdays      Nb. of days to keep .nmon perf. file  10 days
[043] sa.sadm_rch_keepdays       Nb. days to keep unmodified .rch file  35 days
[044] sa.sadm_log_keepdays       Nb. days to keep unmodified .log file  35 days
[045] sa.sadm_max_rchline         Trim rch file to this max. of lines    30 lines
[046] sa.sadm_max_logline        Trim log to this maximum of lines      500 lines
[047] sa.sadm_days_history       Days to keep alert in History file     100 days
[048] sa.sadm_max_arc_line       Max. lines in alert in History file    1000 lines

----- Subnet Network Scanner Section -----
[049] sa.sadm_network1          Network/Netmask 1 inv. IP/Name/Mac     192.168.1.0/24
[050] sa.sadm_network2          Network/Netmask 2 inv. IP/Name/Mac     None
[051] sa.sadm_network3          Network/Netmask 3 inv. IP/Name/Mac     None
[052] sa.sadm_network4          Network/Netmask 4 inv. IP/Name/Mac     None
[053] sa.sadm_network5          Network/Netmask 5 inv. IP/Name/Mac     None

----- ReaR Backup Section -----
[054] sa.sadm_rear_nfs_server    ReaR NFS Server IP or Name           lestrade.maison.ca
[055] sa.sadm_rear_nfs_server_ver ReaR NFS Server Version              3
[056] sa.sadm_rear_nfs_mount_point ReaR NFS Mount Point                /backup/backup_rear
[057] sa.sadm_rear_backup_to_keep ReaR NFS Backup - Nb. to keep        2 copies
[058] sa.sadm_rear_backup_dif     Alert if pct between cur. & prev. size 25 %
[059] sa.sadm_rear_backup_interval Alert if no backup for more than X days 14 days
[060] sa.sadm_rear_backup_script  ReaR backup script name              sadm_rear_backup.sh
[061] sa.sadm_rear_del_failed_backup Del backup if failed integrity check   Y
[062] sa.sadm_rear_backup_concurrent Number of Concurrent rear backup      1
[063] sa.sadm_rear_batch_mode     Run backup in batch mode (y/N)       N
[064] sa.sadm_rear_batch_startup_time Batch startup time                    00:10
[065] sa.sadm_rear_batch_day2run   0=AnyDay,1=Su,2=Mo,3=We,4=Tu,6=Fr,7=Sa "2,7"
[066] sa.sadm_rear_batch_mth2run  0=AnyMonth or [1,2,,,12] Month to run "0"
[067] sa.sadm_rear_batch_date2run 0=AnyDate, Date to run [1,2...27,28] "0"

----- Backup Section -----
[068] sa.sadm_backup_nfs_server   NFS Backup IP or Server Name         batnas.maison.ca
[069] sa.sadm_backup_nfs_server_ver NFS Backup Server Version             3
[070] sa.sadm_backup_nfs_mount_point NFS Backup Mount Point                /volume1/backup/backup_sadmin
[071] sa.sadm_backup_dif          Alert if pct between cur. & prev. size 40 %
[072] sa.sadm_backup_interval     Alert if no backup for more than X days 1 days
[073] sa.sadm_daily_backup_to_keep Daily Backup to Keep                  2 copies
[074] sa.sadm_weekly_backup_to_keep Weekly Backup to Keep                 1 copies
[075] sa.sadm_monthly_backup_to_keep Monthly Backup to Keep                1 copies
[076] sa.sadm_yearly_backup_to_keep Yearly Backup to keep                 1 copies
[077] sa.sadm_weekly_backup_day    Weekly Backup Day (1=Mon,7=Sun)       5
[078] sa.sadm_monthly_backup_date Monthly Backup Date (1-28)            1
[079] sa.sadm_yearly_backup_month Yearly Backup Month (1-12)            12
[080] sa.sadm_yearly_backup_date  Yearly Backup Date (1-31)            31
[081] sa.sadm_backup_concurrent   Number of Concurrent Backup          1
[082] sa.sadm_backup_batch_mode   Run backup in batch mode (y/N)       N
[083] sa.sadm_backup_batch_startup_time Batch startup time                    00:05
[084] sa.sadm_backup_batch_day2run 0=AnyDay,1=Su,2=Mo,3=We,4=Tu,6=Fr,7=Sa "2,7"
[085] sa.sadm_backup_batch_mth2run 0=AnyMonth or [1,2,,,12] Month to run "0"

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[086] sa.sadm_backup_batch_date2run      0=AnyDate, Date to run [1,2...27,28]    "0"

----- Virtual Machines Export Section -----
[087] sa.sadm_vm_export_nfs_server        NFS Export Server                batnas.maison.ca
[088] sa.sadm_vm_export_nfs_server_ver     NFS Export Server Version        3
[089] sa.sadm_vm_export_mount_point        NFS Export Mount Point           /volume1/backup/backup_vm/vbox_exports
[090] sa.sadm_vm_export_dif                 Alert if pct between cur. & prev. size 25 %
[091] sa.sadm_vm_export_to_keep            Nb. of export to keep            2 copies
[092] sa.sadm_vm_export_interval           Alert if no export for more than X days 21 days
[093] sa.sadm_vm_export_alert              Issue an alert if interval reached  Y
[094] sa.sadm_vm_user                     User part of 'vboxusers' user group  jacques
[095] sa.sadm_vm_stop_timeout              Max. seconds given for acpi shutdown  60 seconds
[096] sa.sadm_vm_start_interval            Sec. to sleep between each VM start   30 seconds
[097] sa.sadm_vm_export_concurrent         Number of Concurrent Export         1
[098] sa.sadm_vm_export_batch_mode         Run export in batch mode (y/N)       N
[099] sa.sadm_vm_export_batch_start_time    Start time for launch batch export    00:01
[100] sa.sadm_vm_export_batch_day2run       0=AnyDay, 1=Su, 2=Mo, 3=We, 4=Tu, 6=Fr, 7=Sa "2,7"
[101] sa.sadm_vm_export_batch_mth2run       0=AnyMonth or [1,2,,,12] Month to run "0"
[102] sa.sadm_vm_export_batch_date2run      0=AnyDate, Date to run [1,2...27,28]    "0"

----- System Update Section -----
[103] sa.sadm_osupdate_script              System Update Script Name          sadm_osupdate.sh
[104] sa.sadm_osupdate_interval            Alert if no update for more than X days 15 days
[105] sa.sadm_osupdate_autoremove          Auto remove 'orphaned' packages.    N
[106] sa.sadm_osupdate_flatpak             Update Flatpak packages             N
[107] sa.sadm_osupdate_snap                Update Snap packages                N
[108] sa.sadm_osupdate_reboot_needed        Reboot after each update            N
[109] sa.sadm_osupdate_reboot_time          Sec. given to bring up O/S & app.    240 seconds
[110] sa.sadm_osupdate_lock                 Lock system during the update        Y
[111] sa.sadm_osupdate_concurrent           Number of Concurrent System Updates  1
[112] sa.sadm_osupdate_batch_mode           Run O/S Update in Batch Mode (y/N)   N
[113] sa.sadm_osupdate_batch_start_time     Batch Update Start Time             00:05
[114] sa.sadm_osupdate_batch_day2run       0=AnyDay, 1=Su, 2=Mo, 3=We, 4=Tu, 6=Fr, 7=Sa "2,7"
[115] sa.sadm_osupdate_batch_mth2run       0=AnyMonth or [1,2,,,12] Month to run "0"
[116] sa.sadm_osupdate_batch_date2run      0=AnyDate, Date to run [1,2...27,28]    "0"

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5) Directories variables available to Developers

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[001] sa.dir_base                        SADMIN Root Directory              /opt/sadmin
[002] sa.dir_bin                        SADMIN Scripts Directory            /opt/sadmin/bin
[003] sa.dir_tmp                        System Temporary file(s) Directory /opt/sadmin/tmp
[004] sa.dir_lib                        SADMIN Shell & Python Library Dir. /opt/sadmin/lib
[005] sa.dir_log                        SADMIN Script Log Directory         /opt/sadmin/log
[006] sa.dir_cfg                        SADMIN Configuration Directory      /opt/sadmin/cfg
[007] sa.dir_sys                        System Startup/Shutdown Script Dir. /opt/sadmin/sys
[008] sa.dir_doc                        SADMIN Documentation Directory      /opt/sadmin/doc
[009] sa.dir_pkg                        SADMIN Packages Directory           /opt/sadmin/pkg
[010] sa.dir_dat                        System Data Directory               /opt/sadmin/dat
[011] sa.dir_nmon                       System 'nmon'- Data Collected Dir. /opt/sadmin/dat/nmon
[012] sa.dir_dr                        System Disaster Recovery Info Dir.  /opt/sadmin/dat/dr

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[013]	sa.dir_rch	System Return Code History Dir.	/opt/sadmin/dat/rch
[014]	sa.dir_net	System Network Information Dir.	/opt/sadmin/dat/net
[015]	sa.dir_rpt	System Monitor Report Directory	/opt/sadmin/dat/rpt
[016]	sa.dir_dbb	SADMIN Database Backup Directory	/opt/sadmin/dat/dbb
[017]	sa.dir_setup	SADMIN Setup Directory.	/opt/sadmin/setup
[018]	sa.dir_usr	User root directory	/opt/sadmin/usr
[019]	sa.dir_usr_bin	User Scripts Directory	/opt/sadmin/usr/bin
[020]	sa.dir_usr_lib	User Library Directory	/opt/sadmin/usr/lib
[021]	sa.dir_usr_doc	User Documentation Dir.	/opt/sadmin/usr/doc
[022]	sa.dir_usr_mon	User System Monitor Scripts	/opt/sadmin/usr/mon
[023]	sa.dir_www	SADMIN Web Site Root Directory	/opt/sadmin/www
[024]	sa.dir_www_doc	SADMIN Web Site Documentation Dir.	/opt/sadmin/www/doc
[025]	sa.dir_www_dat	SADMIN Web Site Global Data Dir.	/opt/sadmin/www/dat
[026]	sa.dir_www_lib	SADMIN Web Site PHP Library Dir.	/opt/sadmin/www/lib
[027]	sa.dir_www_tmp	SADMIN Web Temp Working Dir.	/opt/sadmin/www/tmp
[028]	sa.dir_www_perf	SADMIN Web Performance Graph Dir.	/opt/sadmin/www/tmp/perf

6) SADMIN File names use and set by the Python library, for you to use

----- Available File Name Aliases Variables -----

[001]	sa.pid_file	Current script PID file	/opt/sadmin/tmp/sadmlib_std_demo.pid
[002]	sa.cfg_file	SADMIN Main Configuration file	/opt/sadmin/cfg/sadmin.cfg
[003]	sa.cfg_hidden	SADMIN configuration template file	/opt/sadmin/cfg/.sadmin.cfg
[004]	sa.alert_file	Alert Group definition file	/opt/sadmin/cfg/alert_group.cfg
[005]	sa.alert_init	Alert Group definition template file	/opt/sadmin/cfg/.alert_group.cfg
[006]	sa.alert_hist	Alert History file	/opt/sadmin/cfg/alert_history.txt
[007]	sa.alert_hini	Alert History template file	/opt/sadmin/cfg/.alert_history.txt
[008]	sa.tmp_file1	User usable Temp Work File 1	/opt/sadmin/tmp/tmp_file_1.3238021
[009]	sa.tmp_file2	User usable Temp Work File 2	/opt/sadmin/tmp/tmp_file_2.3238021
[010]	sa.tmp_file3	User usable Temp Work File 3	/opt/sadmin/tmp/tmp_file_3.3238021
[011]	sa.log_file	Log File Name	/opt/sadmin/log/sherlock_sadmlib_std_demo.log
[012]	sa.rch_file	Return Code History File Name	/opt/sadmin/dat/rch/sherlock_sadmlib_std_demo.rch
[013]	sa.dbpass_file	SADMIN Database User Password File	/opt/sadmin/cfg/.dbpass
[014]	sa.rpt_file	[SYS]tem [MON]itor report file	/opt/sadmin/dat/rpt/sherlock.rpt
[015]	sa.backup_list	Backup file list	/opt/sadmin/cfg/backup_list.txt
[016]	sa.backup_list_init	Backup file template file	/opt/sadmin/cfg/.backup_list.txt
[017]	sa.backup_exclude	Backup exclude list file name	/opt/sadmin/cfg/backup_exclude.txt
[018]	sa.backup_exclude_init	Backup exclude template file	/opt/sadmin/cfg/.backup_exclude.txt

7) Command Path set and use by SADMIN Python Library for more security.

[001]	sa.cmd_dmidecode	Cmd. 'dmidecode', Get model & type	/usr/sbin/dmidecode
[002]	sa.cmd_bc	Cmd. 'bc', Do some Math.	/usr/bin/bc
[003]	sa.cmd_fdisk	Cmd. 'fdisk', Get Partition Info	/usr/sbin/fdisk
[004]	sa.cmd_which	Cmd. 'which', Get Command location	/usr/bin/which
[005]	sa.cmd_perl	Cmd. 'perl', epoch time Calc.	/usr/bin/perl
[006]	sa.cmd_mutt	Cmd. 'mutt', Used to Send Email	/usr/bin/mutt

[007]	sa.cmd_curl	Cmd. 'curl', To send alert to Slack	/usr/bin/curl
[008]	sa.cmd_lscpu	Cmd. 'lscpu', Socket & thread info	/usr/bin/lscpu
[009]	sa.cmd_nmon	Cmd. 'nmon', Collect Perf Statistic	/usr/bin/nmon
[010]	sa.cmd_parted	Cmd. 'parted', Get Disk Real Size	/usr/sbin/parted
[011]	sa.cmd_ethtool	Cmd. 'ethtool', Get System IP Info	/usr/sbin/ethtool
[012]	sa.cmd_rrdtool	Cmd. 'rrdtool' to produce graph	/usr/bin/rrdtool
[013]	sa.cmd_lsb_release	Cmd. 'lsb_release' to get dist. info	/usr/bin/lsb_release
[014]	sa.cmd_inxi	Cmd. 'inxi' binary location	/usr/bin/inxi
[015]	sa.cmd_ssh	Cmd. 'ssh', SSH to SADMIN client	/usr/bin/ssh

8) Database Information

[001]	sa.db_used	Program need to access Database ?	True
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Show SADMIN Tables:

Tables_in_sadmin
pma__bookmark
pma__central_columns
pma__column_info
pma__designer_settings
pma__export_templates
pma__favorite
pma__history
pma__navigationhiding
pma__pdf_pages
pma__recent
pma__relation
pma__savedsearches
pma__table_coords
pma__table_info
pma__table_uiprefs
pma__tracking
pma__userconfig
pma__usergroups
pma__users
script
script_rch
server
server_category
server_group
server_network

Category Table:

Field	Type	Null	Key	Default	Extra
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cat_id	int(11)	NO	PRI	NULL	auto_increment on update current_timestamp()
cat_code	varchar(10)	NO	UNI	NULL	
cat_desc	varchar(25)	NO		NULL	
cat_active	tinyint(1)	NO		1	
cat_date	timestamp	NO		0000-00-00 00:00:00	
cat_default	tinyint(1)	NO		0	

Table	Non_unique	Key_name	Seq_in_index	Column_name	Collation	Cardinality	Sub_part	Packed	Null	Index
type	Comment	Index_comment								
server_category	0	PRIMARY	1	cat_id	A	8	NULL	NULL		BTREE
server_category	0	key_cat_code	1	cat_code	A	8	NULL	NULL		BTREE

cat_id	cat_code	cat_desc	cat_active	cat_date	cat_default
13	Laptop	Laptop Workstation	1	2023-10-26 12:58:43	0
15	SBC	Raspberry Pi SBC	1	2023-10-21 11:25:49	0
17	Desktop	Workstation	1	2023-10-21 11:23:25	0
25	VM	Virtualize Systems	1	2023-12-31 11:21:58	0
28	Dell	Dell Systems	1	2023-10-26 12:59:19	0
29	HP	HP Systems	1	2023-10-26 12:59:38	0
30	IBM	IBM Systems	1	2023-10-26 12:59:05	0
31	Lenovo	Lenovo Systems	1	2023-10-26 12:59:54	0
32	Server	Home Server	1	2023-12-31 11:21:35	1

Group Table:

Field	Type	Null	Key	Default	Extra
grp_id	int(11)	NO	PRI	NULL	auto_increment on update current_timestamp()
grp_code	varchar(10)	NO	UNI	NULL	
grp_desc	varchar(25)	NO		NULL	
grp_active	tinyint(1)	NO		1	
grp_date	timestamp	NO		current_timestamp()	
grp_default	tinyint(1)	NO		0	

Table	Non_unique	Key_name	Seq_in_index	Column_name	Collation	Cardinality	Sub_part	Packed	Null	Index_type
Comment	Index_comment									
server_group	0	PRIMARY	1	grp_id	A	9	NULL	NULL		BTREE

server_group	0	idx_grp_code	1	grp_code	A	9	NULL	NULL	BTREE
grp_id	grp_code	grp_desc	grp_active	grp_date	grp_default				
4	Retired	System not in use	1	2023-04-05 11:26:50	0				
6	Regular	Application system	1	2023-10-21 11:15:00	0				
9	Dev	Development Systems Group	1	2023-10-21 11:16:20	0				
10	Prod	Production Systems Group	1	2023-10-21 11:16:34	1				
13	Int	Integration Systems Group	1	2023-10-21 11:20:19	0				
16	Acc	Acceptance Systems Group	1	2023-10-22 16:38:23	0				
17	Poc	Proof of concept Systems	1	2023-10-21 15:06:48	0				
19	PreProd	Pre-Production Group	1	2023-10-22 16:39:19	0				
20	Temp	Temporary Systems Group	1	2023-10-26 11:44:28	0				

Server Table:

Field	Type	Null	Key	Default	Extra
srv_id	int(11)	NO	PRI	NULL	auto_increment
srv_name	varchar(15)	NO	UNI	NULL	
srv_domain	varchar(30)	NO		NULL	
srv_desc	varchar(30)	NO		NULL	
srv_tag	varchar(15)	YES		NULL	
srv_note	varchar(30)	YES		NULL	
srv_active	tinyint(1)	NO		1	
srv_sporadic	tinyint(1)	NO		0	
srv_monitor	tinyint(1)	NO		1	
srv_alert_group	varchar(15)	NO		default	
srv_graph	tinyint(1)	NO		1	
srv_cat	varchar(10)	NO		NULL	
srv_group	varchar(10)	NO		NULL	
srv_vm	tinyint(1)	NO		0	
srv_osname	varchar(15)	YES		NULL	
srv_ostype	varchar(10)	NO		linux	
srv_oscodename	varchar(15)	YES		NULL	
srv_osversion	varchar(12)	YES		NULL	
srv_osver_major	varchar(10)	YES		NULL	
srv_date_creation	datetime	YES		0000-00-00 00:00:00	
srv_date_edit	datetime	YES		0000-00-00 00:00:00	
srv_date_update	datetime	YES		0000-00-00 00:00:00	
srv_date_osupdate	datetime	YES		0000-00-00 00:00:00	
srv_kernel_version	varchar(40)	YES		NULL	
srv_kernel_bitmode	smallint(11)	NO		64	
srv_hwd_bitmode	smallint(11)	NO		64	
srv_model	varchar(40)	YES		NULL	
srv_serial	varchar(25)	YES		NULL	
srv_memory	int(11)	YES		NULL	

[illegible]

srv_export_dow	varchar(8)	NO		YNNNNNNN								
srv_export_hrs	smallint(6)	NO		3								
srv_export_min	smallint(6)	NO		15								
srv_vm_version	varchar(15)	NO		NULL								
srv_vm_host	varchar(25)	YES		NULL								
srv_vm_start_seq	smallint(11)	NO		0								
srv_update_date	date	NO		current_timestamp()								
srv_update_autorm	tinyint(1)	NO		0								
srv_update_flatpak	tinyint(1)	NO		0								
srv_update_snap	tinyint(1)	NO		0								
srv_update_seq	mediumint(9)	NO		0								
srv_update_maxdays	smallint(6)	YES		0								
srv_update_block	varchar(1)	NO		N								
srv_backup_date	date	YES		NULL								
srv_backup_nostat	tinyint(1)	NO		0								
srv_backup_seq	int(9)	NO		0								
srv_backup_block	varchar(1)	NO		N								
srv_export_seq	mediumint(9)	NO		0								
srv_export_block	varchar(1)	NO		N								
srv_export_compaq	tinyint(1)	NO		0								
srv_img_seq	mediumint(9)	NO		0								
srv_img_rmfail	tinyint(1)	NO		0								
srv_img_block	varchar(1)	NO		N								
srv_system_lock	tinyint(4)	NO		0								
srv_warning_group	varchar(15)	NO		default								
srv_info_group	varchar(15)	NO		default								
srv_email_startup	tinyint(1)	NO		0								
srv_email_shutdown	tinyint(1)	NO		0								

Table	Non_unique	Key_name	Seq_in_index	Column_name	Collation	Cardinality	Sub_part	Packed	Null	Index_type	C
omment	Index_comment										
server	0	PRIMARY	1	srv_id	A	28	NULL	NULL		BTREE	
server	0	idx_srv_name	1	srv_name	A	28	NULL	NULL		BTREE	

Script table:

Field	Type	Null	Key	Default	Extra
scr_id	mediumint(8) unsigned	NO	PRI	NULL	auto_increment
scr_name	varchar(40)	YES	UNI	NULL	
scr_desc	varchar(80)	YES		NULL	
scr_version	varchar(10)	NO		NULL	
scr_alert_grp	varchar(15)	YES		NULL	
scr_alert_type	smallint(6)	NO		1	

scr_sysadm_email	varchar(50)	NO		NULL		
scr_multiple_exec	tinyint(1)	NO		0		
scr_use_rch	tinyint(1)	NO		1		
scr_log_type	varchar(1)	NO		B		
scr_log_append	tinyint(1)	NO		0		
scr_log_header	tinyint(1)	NO		1		
scr_log_footer	tinyint(1)	NO		1		
scr_active	tinyint(1)	NO		1		
scr_max_logline	int(11)	NO		500		
scr_max_rchline	int(11)	NO		35		
scr_pid_timeout	int(11)	NO		7200		
src_max_inactive_days	int(3)	NO		31		
scr_root_exec	tinyint(4)	NO		0		
scr_run_sadmin	tinyint(4)	NO		0		
scr_os	varchar(10)	NO		any		
scr_last_activity	datetime	NO		current_timestamp()		
scr_severity	smallint(6)	NO		1		
scr_exclude	set('0','1')	NO		0		
scr_log_keepdays	smallint(6)	NO		0		
scr_rch_keepdays	smallint(6)	NO		0		
scr_avg_exec	int(11)	YES		0		
scr_debug	tinyint(1)	NO		0		
scr_grp_warning	varchar(15)	NO		NULL		
scr_grp_info	varchar(15)	NO		NULL		
scr_grp_error	varchar(15)	NO		NULL		
scr_lock_timeout	int(11)	NO		3600		

Table	Non_unique	Key_name	Seq_in_index	Column_name	Collation	Cardinality	Sub_part	Packed	Null	Index_type	C
omment	Index_comment										
script	0	PRIMARY	1	scr_id	A	0	NULL	NULL		BTREE	
script	0	scr_name_key	1	scr_name	A	0	NULL	NULL	YES	BTREE	

Script_rch:

Field	Type	Null	Key	Default	Extra
scrdet_host	varchar(15)	NO	PRI	NULL	
scrdet_name	varchar(40)	NO	PRI	NULL	
srcdet_start	timestamp	NO	PRI	0000-00-00 00:00:00	
srcdet_end	timestamp	YES		NULL	
scrdet_status	tinyint(2)	NO		0	
scrdet_active	int(1)	NO		1	
srcdet_elapse	time	YES		NULL	

